

WASP-2b

R-1573

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-2_190904 · created 2026-08-01T22:55:05+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458730.713 +/- 0.0036 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.149 +/- 0.024
Transit depth (Rp/Rs)^2	2.23 +/- 0.7 [%]
Orbital Inclination (inc)	84.1 +/- 1.37
Airmass coefficient 1 (a1)	1.018 +/- 0.011
Airmass coefficient 2 (a2)	-0.015 +/- 0.063
Scatter in the residuals of the lightcurve fit is	1.07 %
Best Comparison Star	#1 - [353, 140]
Optimal Method	PSF photometry
Transit Duration (day)	0.061 +/- 0.023

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.149 ± 0.024 vs 0.1326 (deviation 0.61σ)
Duration fit vs published	0.061 d vs 0.0737 d (deviation 0.49σ)
Mid-time O–C	7.0 min
Depth SNR	5.5
Residual scatter	1.07 %

Light curve

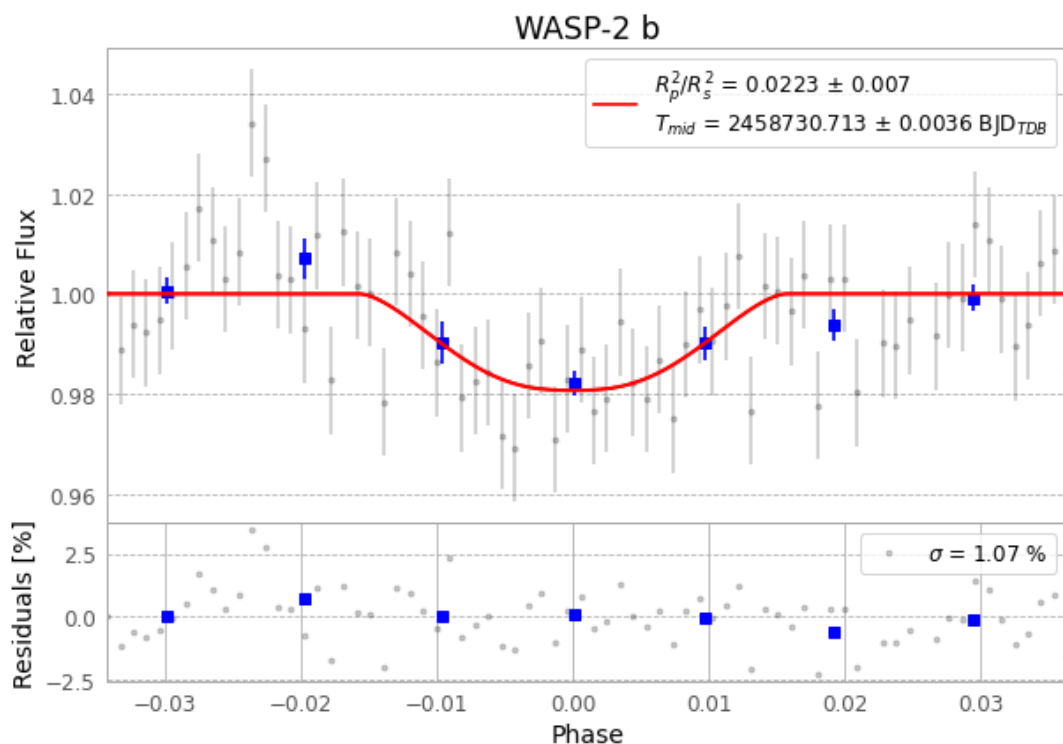


Figure 1 — FinalLightCurve_WASP-2 b_2019-09-03.png (SHA-256 80540c2137b12579...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [159, 199], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 846a4a14eb9ef863...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

74 FITS frames (49 MB), WASP-2190904031212.FITS → WASP-2190904065112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-190904.log (cde3d1b7bd55...), FinalLightCurve_WASP-2 b_2019-09-03.png (80540c2137b1...), FinalLightCurve_WASP-2 b_2019-09-03.csv (9ac5adcc80ed...)

Compute resources

Wall time 170.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 22:55Z.